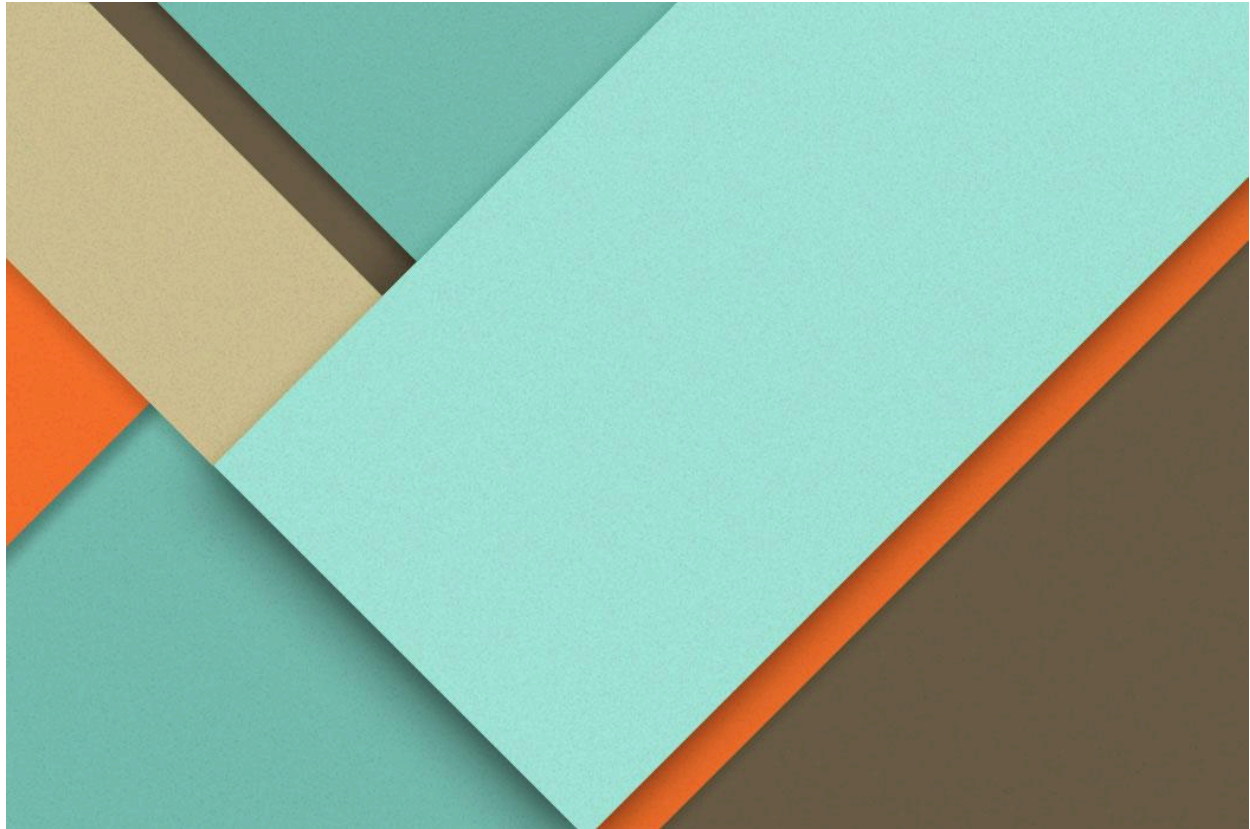




T.A.N. Solutions

08.15.2026

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BSIT 4D

CCS Building

Laguna State Polytechnic University - Santa Cruz Campus

Nature of Business

A software development startup delivering custom web and mobile applications, internal tools, and cloud-based solutions for SMEs.

Company Vision

To empower businesses with reliable, secure, and scalable software that accelerates growth and operational efficiency.

Company Mission

To design, build, and maintain high-quality software products using modern development practices, cloud infrastructure, and strong security controls—while fostering a collaborative, learning-driven team culture.

Organizational Structure

- CEO/Founder
- IT Manager (oversees infrastructure and security)
- Departments: IT, Human Resources, Finance, Sales
- Junior System Administrator (you) reports to IT Manager

Employee Distribution

Department	Employees
Information Technology	5
Human Resources	4
Finance	5
Sales	6
Total	20

Part 2 – Enterprise Hardware Inventory

Asset ID	Hardware	Quantity	Department / Use	Purpose & Justification
HW-001	Desktop Computers	12	IT (3), Finance (5), HR (2), Sales (2)	Cost-effective, stable workstations for roles that primarily work from fixed desks.
HW-002	Laptops	8	IT (2), Sales (4), HR (2)	Mobility for sales and hybrid staff; IT staff need laptops for on-site support and testing.
HW-003	Server	1	IT (shared)	Hosts domain services (AD DS), file shares, internal apps, and backup repository.
HW-004	Router	1	IT (network core)	Routes traffic between LAN, WAN, and VLANs; supports QoS and remote access.
HW-005	Switch	1	IT (network core)	Connects all wired devices; supports VLANs for departmental segmentation.
HW-006	Network Printer	1	Shared (HR/Finance heavy use)	Centralized printing/scanning; reduces per-device cost and simplifies driver management.
HW-007	UPS	2	Server room / network rack	Protects server and network gear from power surges and allows graceful shutdown.
HW-008	Wireless Access Point	2	All departments	Provides reliable, secure Wi-Fi coverage across the single-floor office.
HW-009	NAS Storage	1	IT (shared)	Central file storage with redundancy; supports departmental shares and versioning.
HW-010	External Backup Drive	2	IT (backup rotation)	Offline/air-gapped backup copies for disaster recovery.
HW-011	Monitors	14	Desktop users + some laptop docking	Dual-monitor productivity for finance/IT; one extra as spare.

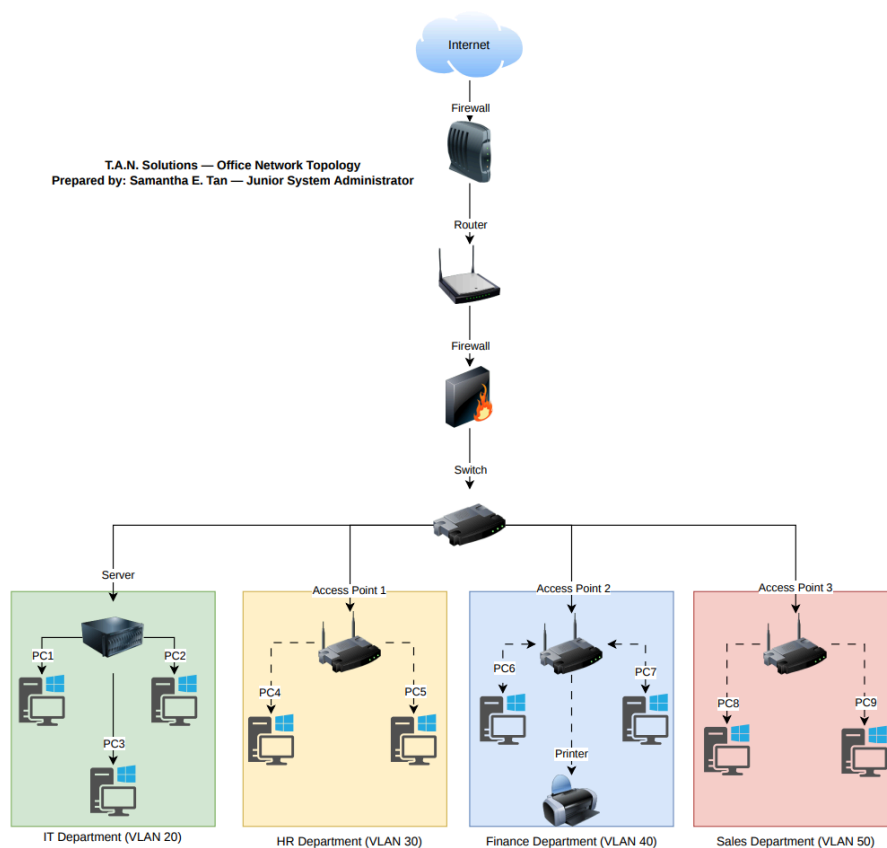
Part 3 – Enterprise Software Inventory

Software	Version	License	Purpose & Why Needed
Windows 11 Pro	22H2+	Commercial	Standard OS for desktops/laptops; BitLocker, Group Policy, and domain join support.
Ubuntu Server	24.04 LTS	Open source	Stable server OS for internal services
Microsoft Office	Current	Subscription	Productivity suite plus OneDrive/Teams collaboration.
VS Code	Latest	Free	Primary code editor for developers; extensions for multiple languages.
Git	Latest	Open source	Version control for source code; essential for development workflows.
GitHub Desktop	Latest	Free	Simplifies Git operations for non-CLI users; integrates with GitHub/GitLab.
VirtualBox	Latest	Open source	Local virtualization for testing environments, labs, and safe software trials.
Google Chrome	Latest	Free	Standard browser for web apps, testing, and SaaS tools.
Microsoft Defender	Built-in + cloud	Included with M365/Windows	Baseline antivirus/EDR; integrates with Intune/Defender portal if used.
AnyDesk	Latest	Free/Pro	Remote support tool for internal troubleshooting and quick assistance.
7-Zip	Latest	Open source	File compression/archiving for backups, logs, and large transfers.
Figma	Latest	Free/Freemium	Developer and design tooling commonly used in software startups.

Part 4 – Enterprise Network Inventory

Item	Quantity	Specification	Purpose
ISP Modem	1	Fiber broadband	Primary internet handoff from ISP.
Router	1	Dual-WAN capable, VLAN support, VPN	Routes traffic, enforces policies, supports site-to-site/remote access.
Next-Gen Firewall	1	UTM features	Protects perimeter; segments traffic; inspects threats.
Switch	1	24-port Gigabit, PoE+, VLAN, L2/L3 lite	Connects wired devices; powers APs; segments departments via VLANs.
Wireless Access Point	2	Wi-Fi 6, PoE, WPA3-Enterprise ready	Provides secure, high-capacity Wi-Fi across the floor.
Patch Panel	1	24-port CAT6	Terminates wall jacks; keeps cabling organized in rack.
CAT6 Cables	~30–40 runs	1–30m assorted lengths	Wired connections for desktops, APs, printer, server, NAS.
RJ45 Connectors	50+	CAT6-rated	For patch cables and wall outlets.
Spare SFP / Cables	1 set	As needed	For future uplinks or redundancy.

Part 5 – Enterprise Network Diagram



Part 6 – System Administration Roles

Helpdesk Technician

- **Responsibilities:** First-line support, process IT support requests, software installs, basic hardware troubleshooting, user onboarding/offboarding.
- **Skills:** Communication, patience, OS familiarity, remote support tools, documentation.
- **Common Tools:** Ticketing system, remote tools, M365 Admin, AD user mgmt.
- **Certifications:** CompTIA A+, ITIL 4 Foundation, Microsoft 365 Certified: Fundamentals.

Network Administrator

- **Responsibilities:** Design/maintain LAN/WLAN, VLANs, DHCP/DNS, firewall rules, monitoring, performance tuning, change management.
- **Skills:** TCP/IP, routing/switching, Wi-Fi design, security basics, scripting.
- **Common Tools:** Managed switches/AP controllers, firewall UI, Wireshark, ping/traceroute, NMS.
- **Certifications:** CompTIA Network+, Cisco CCNA, Juniper JNCIA, Fortinet NSE 4.

Linux System Administrator

- **Responsibilities:** Install/configure Linux servers, user/permission mgmt, patching, services, backups, monitoring, hardening.
- **Skills:** Bash scripting, package mgmt, systemd, storage, networking, security.
- **Common Tools:** SSH, rsync, cron, Ansible, Nagios/Zabbix, Git.
- **Certifications:** Linux+, RHCSA/RHCE, LFCS.

Cloud Administrator

- **Responsibilities:** Provision and manage cloud resources, IAM and policies, cost monitoring, backups, security baselines.
- **Skills:** Cloud platforms, IaC basics, networking in cloud, monitoring/logging.
- **Common Tools:** AWS Console/CLI, Azure Portal, Terraform, CloudWatch/Monitor, S3/Blob storage.
- **Certifications:** AWS Certified SysOps Administrator / Solutions Architect Associate, Microsoft Azure Administrator, Google Cloud Associate Cloud Engineer.

How they work together

Helpdesk fixes user issues and escalates bigger problems. Network Admin keeps connections reliable and secure. Linux Admin keeps servers and core services stable and updated. Cloud Admin adds cloud capacity and enforces security policies. Together, they keep systems secure, up, and scalable.

Part 7 – Infrastructure Recommendations

Internet Provider

- Choose a fiber business plan with at least 300–500 Mbps symmetric speeds and a static IP option.
- Reason: Supports cloud dev tools, large repos, video calls, and SaaS without bottlenecks; symmetric speeds help uploads.

Server Specifications

- CPU: 8–12 cores
- RAM: 32–64 GB ECC
- Storage: 2× SSD for OS/apps + 4× HDD/SSD in RAID for data/NAS
- NIC: Dual 1GbE
- Reason: Handles AD, file services, internal apps, and monitoring with headroom for growth.

Backup Strategy (3-2-1)

- 3 copies of data
- 2 different media
- 1 offsite
- Schedule: Daily incremental, weekly full; test restores quarterly.
- Reason: Protects against ransomware, hardware failure, and human error.

Security Recommendations

- Deploy NGFW with IPS, web filtering, and application control.
- Enforce WPA3-Enterprise or WPA2-Enterprise with 802.1X for Wi-Fi.
- Segment networks via VLANs.
- Enable endpoint protection with cloud management; consider EDR if budget allows.
- Reason: Reduces attack surface and contains incidents.

Antivirus / Endpoint Protection

- Use Microsoft Defender for Endpoint with cloud-delivered protection and automatic updates.
- Add browser isolation policies and controlled folder access for ransomware protection.
- Reason: Strong baseline protection integrated with Windows and M365.

Password Policy

- Minimum 12 characters; complexity not overly strict but avoid common words.
- Enforce MFA for all admin and cloud accounts.

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- Password manager company-wide; rotate only on suspicion of compromise.
 - Lockout after 5 failed attempts; 15-minute idle timeout on workstations.
 - Reason: Balances security and usability while preventing credential attacks.

Expansion Plan (12–24 months)

- Add a second ISP line for failover when headcount reaches ~30.
- Upgrade switch to stackable or add a second 24-port switch for redundancy.
- Introduce a dedicated backup server or cloud backup service.
- Formalize IT policies.
- Reason: Ensures continuity and scalability as the startup grows.

Part 8 – Personal Reflection

This project made me realize that building IT infrastructure isn't just about buying computers and plugging them in. The real work starts with understanding how the business actually operates. Once I mapped out the departments and their needs, it became easier to justify why we need specific devices, how many ports the switch should have, or why VLANs matter. I started thinking like a sysadmin: not just “what do we need?” but “why, how will it be used, and what happens if it breaks?”

The hardest part was the network diagram. I kept second-guessing myself—how many VLANs are enough, where should the firewall sit, will two access points really cover the whole office? Drawing it in Draw.io was also tricky because I had to make sure the symbols were correct, the labels were clear, and the layout wasn't a mess. It forced me to be specific about things I usually skip, like cable types, PoE requirements, and port counts.

Planning first matters because it saves money and prevents headaches. Without a plan, you end up buying the wrong gear, leaving security holes, or building a network that can't grow. A good plan lines up tech with business needs, sets standards, and gives you a realistic budget. For a startup, that's huge—every peso counts. It also gives you a reference when things go wrong, because you know what's supposed to be there and how it should behave.

This project helped me grow because I got to practice the full sysadmin workflow: gathering requirements, making inventories, designing the network, writing policies, and explaining everything clearly. I also got better at documenting my decisions, which is something I'll need in real jobs. Most of all, I now see infrastructure as something that needs constant care—documented, monitored, and improved. That “plan first, then execute” mindset is what I'll take forward in my career.